Jesús Alejandro Alzate Grisales

AI Researcher and Data Scientist

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PROFILE

AI researcher and data scientist with 5+ years of experience turning real-world data into validated machine learning models: 3D and 4D medical images, clinical text and tabular data. I work across the full research cycle, from hypothesis and data curation to training, evaluation and prototypes that clinicians can use. Author of 30 publications (7 as first author, 349 citations), with deep learning models validated across 16 hospitals. PhD candidate at Universitat Politècnica de València with a competitive AECC research grant.

SKILLS

Machine learning
Deep learning, computer vision, multimodal AI, CNNs, transformers, foundation models, transfer learning, ensembles, federated learning, interpretability

Programming
Python, PyTorch, TensorFlow, MONAI, nnU-Net, NumPy, Pandas, scikit-learn, C++, MATLAB, SQL, Bash

Engineering
Prototypes and MVPs, MLOps, Docker, Flask, Git, Linux, HPC clusters

LLMs and NLP
LLM-based agents and workflows, prompting and few-shot retrieval, fine-tuning BERT models, information extraction, de-identification of clinical text

Imaging and data
3D and 4D volumetric data, segmentation, registration, denoising, super-resolution, radiomics, statistics, DICOM, 3D Slicer, OHIF

Ways of working
Research design, technical coordination, multidisciplinary teams with clinicians and engineers, teaching and scientific writing

EXPERIENCE

Data Scientist, AI Research
10/2022 – Present

FISABIO-CIPF Joint Unit for Biomedical Imaging and AI, Valencia, Spain

- Develop and evaluate deep learning models for segmentation and classification of MRI and CT volumes with PyTorch and MONAI on an HPC cluster.
- Built a prostate cancer detection model trained on 4,700 MRI sessions from 16 centres plus a public dataset; transfer learning and model stacking reached AUC 0.816 on an independent multi-centre test set (Scientific Reports, 2026).
- Adapted the Segment Anything foundation model to 3D lesion segmentation (SAM-UNETR, IEEE Access, 2023).
- Compared fine-tuned BERT and open LLMs (Gemma, Qwen) to de-identify Spanish clinical text (DisMed-LLM, first author, 2026); now developing DisMed-Agents, an agent-based toolkit with critic, validator, specialist and consensus strategies, packaged with a CLI, web app and Docker.
- Use LLMs to extract structured clinical variables from free-text medical reports.
- Combine imaging, clinical variables and radiomic features in multimodal models; validated model explanations against expert annotations, which matched in 88% of cases.
- Technical coordinator of ARTEMISA (AI and augmented reality for neurosurgery): turned research prototypes into tools in 3D Slicer and OHIF; worked on federated learning.

PhD Research Stay
01/2026 – 04/2026

ARTORG Center for Biomedical Engineering Research, University of Bern, Switzerland

- Designed attention-based CNNs to denoise and super-resolve 4D flow MRI, bringing 3T scans closer to 7T quality, with a full pipeline for registration, vessel segmentation, training and evaluation. Co-first author, CMBE 2026, Kobe.

Data Analyst
12/2021 – 12/2022

SIGMA Ingeniería, Manizales, Colombia

- Built predictive models, NLP and computer vision solutions for healthcare and urban waste management, from raw data to production; co-authored a simulation-based optimisation model for waste collection (Production, 2024).

Young Researcher in Deep Learning
02/2021 – 02/2022

Minciencias and Universidad Autónoma de Manizales, Colombia

- Trained CNNs for COVID-19 detection in chest X-rays (75 citations); collected, labelled and released the open COV-Caldas dataset (Scientific Data, 2022).

Research Stay, Delfín Programme
2021

Universidad Nacional Autónoma de México (UNAM), Mexico

- Applied BERT language models to classify pragmatic language in psychological reports.

EDUCATION

PhD in Health and Well-being Technologies
10/2024 – Present

Universitat Politècnica de València, Spain

Multimodal AI for the detection and characterisation of prostate cancer (PROST-AID). AECC Predoctoral Grant, 2025–2028.

MSc in Biomedical Engineering
2023 – 2024

Universitat Politècnica de València, Spain

Thesis on deep learning for prostate cancer detection in MRI, graded with highest honours.

BEng in Biomedical Engineering
2017 – 2021

Universidad Autónoma de Manizales, Colombia

Honours in eight consecutive semesters. Teaching assistant in mathematics, signals, image processing and computational models.

SELECTED PUBLICATIONS

Clinically significant prostate cancer detection with deep learning in a multi-center magnetic resonance imaging study. First author. Scientific Reports, 2026.

DisMed-LLM: De-Identifying Spanish Medical Text with Large Language Models. First author. Journal of Healthcare Informatics Research, 2026.

Integrating deep learning and geometry-based microstructural analysis of technical bamboo fibers for bio-composite engineering. Composites Part A, 2026.

From diverse CT scans to generalization: towards robust abdominal organ segmentation. BMC Medical Imaging, 2026.

SAM-UNETR: Clinically Significant Prostate Cancer Segmentation Using Transfer Learning From Large Model. First author. IEEE Access, 2023.

GBRAS-Net: A Convolutional Neural Network Architecture for Spatial Image Steganalysis. IEEE Access, 2021. 98 citations.

Full list of 30 publications on orcid.org/0000-0003-1021-2050

AWARDS

- AECC Predoctoral Grant, Spanish Association Against Cancer, 2025: competitive three-year research funding.
- Young Researcher award, Minciencias (Colombian Ministry of Science), 2021.

TEACHING AND TALKS

- Course content author, UNIR, 2026: clinical data science; NLP and generative models in mental health.
- Instructor, AI for Research course, EVES, Valencia, 2026.
- Talks at CMBE 2026 (Kobe), EAU 2025 (Madrid), JBI 2025 (Madrid) and the Spanish National Urology Congress 2024 and 2026.

LANGUAGES

Spanish (native), English (B2, professional)